1. What is force view? What is inline view?

- **Force View:** A view that is created regardless of whether the underlying base tables or views exist. It is useful when creating a view referencing objects that do not yet exist.
- **Inline View:** A subquery in the FROM clause that behaves like a temporary table. It's used to simplify complex queries or provide intermediate results.

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2. What is Oracle Hints?

- Oracle Hints are instructions provided to the optimizer to influence the execution plan of a query. For example, /*+ FULL(table) */ tells Oracle to perform a full table scan on the specified table.

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3. How to update a complex view?

- Use the INSTEAD OF trigger if the complex view involves joins, aggregate functions, or cannot be directly updated.

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4. What are the query optimization techniques in Oracle?

- Techniques include:
    - Using indexes.
    - Avoiding full table scans.
    - Partitioning tables.
    - Optimizing joins and subqueries.
    - Using Oracle Hints.
    - Analyzing execution plans.

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5. What are global temporary tables?

- Temporary tables whose data is session-specific but structure is shared across sessions. Data is either transaction-specific or session-specific based on the table definition.

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6. Difference between UNION and UNION ALL?

- **UNION:** Removes duplicates, leading to slower performance.
- **UNION ALL:** Includes duplicates, faster performance.

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7. Explain candidate key, primary key, and super key?

- **Candidate Key:** A minimal set of attributes that uniquely identify a row.
- **Primary Key:** A candidate key chosen as the unique identifier for a table.
- **Super Key:** A superset of a candidate key.

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8. Query to display the employee whose name starts with 'S'?

```
SELECT * FROM employees WHERE name LIKE 'S%';
```

---

9. How to delete duplicate records?

```
DELETE FROM employees
WHERE rowid NOT IN (
 SELECT MIN(rowid)
 FROM employees
 GROUP BY column1, column2, ...);
```

---

10. What is materialized view log?

- A schema object that records changes to a base table, used for fast refreshes of a materialized view.

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11. What is index and how does it improve query performance?

- An index is a database object that improves query performance by providing a faster path to locate rows in a table. Examples include B-tree and bitmap indexes.

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12. Query to display the nth highest salary from emp table?

```
SELECT Salary
FROM (
  SELECT DISTINCT Salary
  FROM Employee
  ORDER BY Salary DESC
)
WHERE ROWNUM = n;
```

---

13. What is normalization? Explain 1NF, 2NF?

- **Normalization:** Process of organizing data to reduce redundancy.
    - **1NF:** Ensures atomic values, no repeating groups.
    - **2NF:** Removes partial dependency; all non-key attributes depend on the entire primary key.

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14. Write a query to display the current date?

SELECT SYSDATE FROM DUAL;

---

15. Refresh types in materialized view?

- **Complete Refresh:** Rebuilds the entire view.
- **Fast Refresh:** Applies changes using the materialized view log.
- **Force Refresh:** Automatically chooses between complete or fast.

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16. Difference between clustered and non-clustered index in SQL?

- **Clustered Index:** Stores table data physically in order.
- **Non-clustered Index:** Maintains a separate structure for data pointers.

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17. Difference between DELETE and TRUNCATE?

- **DELETE:** Removes rows based on condition; can be rolled back.
- **TRUNCATE:** Removes all rows; cannot be rolled back.

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18. Explain self-join in SQL and when it is used?

- A join where a table joins itself. Useful for hierarchical data, e.g., finding a manager's employees.

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19. What is Explain Plan?

- A tool to display the execution plan chosen by the optimizer for a query.

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20. Difference between EXISTS and IN?

- **EXISTS:** Efficient for correlated subqueries.
- **IN:** Efficient for static list comparisons.

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21. What are window functions?

- Functions that perform calculations across a result set's rows (e.g., ROW_NUMBER, RANK).
    - Example:
    - SELECT name, salary, RANK() OVER (ORDER BY salary DESC) AS rank FROM employees;

---

22. What are synonyms? Types of synonyms?

- **Synonym:** Alias for a database object.
    - **Public Synonym:** Accessible by all users.
    - **Private Synonym:** Accessible only by the owner.

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23. Difference between DECODE and CASE?

- **DECODE:** Oracle-specific; evaluates expressions.
- **CASE:** ANSI-compliant, more flexible.

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24. Difference between view and materialized view?

- **View:** Virtual table, no storage.
- **Materialized View:** Stores data physically for faster access.

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25. Difference between LEFT OUTER JOIN and RIGHT OUTER JOIN?

- **LEFT OUTER JOIN:** Includes all records from the left table.
- **RIGHT OUTER JOIN:** Includes all records from the right table.

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26. What is CTE in Oracle?

- A Common Table Expression (CTE) is a temporary result set used within a query.

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27. Types of subqueries?

- **Single-row:** Returns one row.
- **Multi-row:** Returns multiple rows.
- **Correlated:** Depends on the outer query.

- **Non-correlated:** Independent subquery.

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28. What is Pivot function?

- Converts rows into columns for better data presentation.

---

29. Query to display first five highest salary employees?

```
SELECT * FROM (
  SELECT name, salary, RANK() OVER (ORDER BY salary DESC) AS rank FROM employees
) WHERE rank <= 5;
```

---

30. What are joins? Types of joins?

- Joins combine rows from two or more tables.
  - Inner, Left Outer, Right Outer, Full Outer, Cross Join.

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31. Set operators in Oracle?

- UNION, UNION ALL, INTERSECT, MINUS.

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32. What is table partitioning in SQL?

- Divides a table into smaller segments for performance optimization.

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33. How to find the list of indexes for a table?

```
SELECT index_name FROM user_indexes WHERE table_name = 'EMPLOYEES';
```

---

34. Query to display employees with max salary per department?

```
SELECT department_id, name, salary FROM employees e1
WHERE salary = (SELECT MAX(salary) FROM employees e2 WHERE e1.department_id =
e2.department_id);
```

---

35. Benefits and drawbacks of indexes?

- **Benefits:** Faster queries, reduced I/O operations.
- **Drawbacks:** Increased storage, slower write operations.

36. What is SUBSTR(), INSTR()?

- **SUBSTR():** Extracts a substring.
- **INSTR():** Finds position of a substring.

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37. Difference between REPLACE and TRANSLATE?

- **REPLACE:** Substitutes a substring with another.
- **TRANSLATE:** Replaces characters individually.

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38. What is Btree, Bitmap, Function-based index?

- **Btree:** Balanced tree structure, efficient for unique values.
- **Bitmap:** Compact, efficient for low-cardinality data.
- **Function-based:** Indexes on expressions or functions.

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39. Null value functions in SQL?

- **NVL(), COALESCE(), NVL2(), ISNULL().**

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40. What is dynamic SQL?

- SQL statements constructed and executed at runtime using EXECUTE IMMEDIATE or DBMS_SQL.
